Table 6–10. Transmission Fluid Temperature Sensor Resistance vs. Temperature *(cont'd)*

Temperature		Resistance		
(°C)	(°F)	Minimum (Ω)	Nominal (Ω)	Maximum (Ω)
150	302	46.08	47.46	48.84
155	311	41.04	42.27	43.50
160	320	36.65	37.74	38.84

Table 6–11. TPS—Distance (mm) of Travel vs. Volts

mm	Volts	mm	Volts	mm	Volts	mm	Volts
0	0	12	1.317	24	2.634	36	3.951
1	0.110	13	1.427	25	2.744	37	4.061
2	0.220	14	1.537	26	2.854	38	4.171
3	0.329	15	1.646	27	2.964	39	4.281
4	0.439	16	1.756	28	3.073	40	4.390
5	0.549	17	1.866	29	3.183	41	4.500
6	0.659	18	1.976	30	3.293	42	4.610
7	0.768	19	2.085	31	3.403	43	4.720
8	0.878	20	2.195	32	3.512	44	4.829
9	0.988	21	2.305	33	3.622	45	4.939
10	1.098	22	2.415	34	3.732	46	5.049
11	1.207	23	2.524	35	3.842		

6–6. DTC LIST AND DESCRIPTION INDEX

Table 6–12. Diagnostic Trouble Codes and Descriptions

DTC	Description	CHECK TRANS Light	Diagnostic Response	Page
P0122	Pedal Position Sensor Circuit—Low Voltage	Yes		6-19
P0123	Pedal Position Sensor Circuit—High Voltage	Yes		6-23
P0218	Transmission Fluid Over Temperature	No		6-27
P0561	System Voltage—Performance	No		6-29
P0562	System Voltage—Low	No		6-33
P0563	System Voltage—High	No		6-36
P0602	TCM Not Programmed	No		6-39
P0603	Internal Control Module Keep Alive Memory Error	Yes		6-40
P0604	Control Module Random Access Memory (RAM)	Yes		6-41

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Table 6–12. Diagnostic Trouble Codes and Descriptions *(cont'd)*

DTC	Description	CHECK TRANS Light	Diagnostic Response	Page
P0607	Control Module Performance	No		6-42
P0614	Torque Control Data Mismatch – ECM/TCM	Yes		6-44
P0634	TCM Internal Temperature Too High	Yes		6-47
P0642	Sensor Reference Voltage A Circuit Low	Yes		6-49
P0643	Sensor Reference Voltage A Circuit High	Yes		6-52
P0657	Actuator Supply Voltage 1 Open	Yes		6-54
P0658	Actuator Supply Voltage 1 (HSD1)—Low	Yes		6-58
P0659	Actuator Supply Voltage 1 (HSD1)—High	Yes		6-62
P0701	Transmission Control System—Performance	No		6-67
P0703	Brake Switch Circuit	No		6-69
P0706	Transmission Range Sensor Circuit—Performance	Yes		6-72
P0708	Transmission Range Sensor Circuit—High Input	No		6-77
P0711	Transmission Fluid Temperature Sensor Circuit—Performance	No		6-82
P0712	Transmission Fluid Temperature Sensor Circuit—Low Input	No		6-87
P0713	Transmission Fluid Temperature Sensor Circuit—High Input	Yes		6-92
P0715	Turbine Shaft Speed Sensor Circuit	Yes		6-97
P0716	Turbine Speed Sensor Circuit—Performance	Yes		6-101
P0717	Turbine Speed Sensor Circuit—No Signal	Yes		6-105
P071A	Neutral At Stop Input Failed On	No		6-109
P0720	Output Shaft Speed Sensor Circuit	Yes		6-112
P0721	Output Speed Sensor Circuit—Performance	Yes		6-116
P0722	Output Speed Sensor Circuit—No Signal	Yes		6-120
P0725	Engine Speed Sensor Circuit	No		6-124
P0726	Engine Speed Sensor Circuit—Performance	No		6-128
P0727	Engine Speed Sensor Circuit—No Signal	No		6-131
P0729	Incorrect 6th Gear Ratio	Yes		6-135

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Table 6–12. Diagnostic Trouble Codes and Descriptions *(cont'd)*

DTC	Description	CHECK TRANS Light	Diagnostic Response	Page
P0731	Incorrect 1st Gear Ratio	Yes		6-138
P0732	Incorrect 2nd Gear Ratio	Yes		6-141
P0733	Incorrect 3rd Gear Ratio	Yes		6-145
P0734	Incorrect 4th Gear Ratio	Yes		6-149
P0735	Incorrect 5th Gear Ratio	Yes		6-153
P0736	Incorrect Reverse Ratio	Yes		6-156
P0741	Torque Converter Clutch (TCC) System—Stuck Off	Yes		6-160
P0742	Torque Converter Clutch (TCC) System—Stuck On	Yes		6-162
P0751	Shift Solenoid 1 (SS1) Valve Performance—Stuck Off	Yes		6-164
P0752	Shift Solenoid 1 (SS1) Valve Performance—Stuck On	Yes		6-170
P0756	Shift Solenoid 2 (SS2) Valve Performance—Stuck Off	Yes		6-176
P0757	Shift Solenoid 2 (SS2) Valve Performance—Stuck On	Yes		6-182
P0761	Shift Solenoid 3 (SS3) Valve Performance—Stuck Off	Yes		6-188
P0762	Shift Solenoid 3 (SS3) Valve Performance—Stuck On	Yes		6-194
P0776	Pressure Control Solenoid 2 (PCS2)—Stuck Off	Yes		6-200
P0777	Pressure Control Solenoid 2 (PCS2)—Stuck On	Yes		6-204
P0837	Four Wheel Drive (4WD) Switch Circuit Range/Performance	Yes		6-208
P0842	Transmission Pressure Switch 1 (PS1) Circuit—Low	Yes		6-211
P0843	Transmission Pressure Switch 1 (PS1) Circuit—High	Yes		6-216
P0847	Transmission Pressure Switch 2 (PS2) Circuit—Low	Yes		6-221
P0848	Transmission Pressure Switch 2 (PS2) Circuit—High	Yes		6-227
P0872	Transmission Pressure Switch 3 (PS3) Circuit—Low	Yes		6-232
P0873	Transmission Pressure Switch 3 (PS3) Circuit—High	Yes		6-237
P0877	Transmission Fluid Pressure Switch 4 (PS4) Circuit—Low	Yes		6-242

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Table 6–12. Diagnostic Trouble Codes and Descriptions *(cont'd)*

DTC	Description	CHECK TRANS Light	Diagnostic Response	Page
P0878	Transmission Fluid Pressure Switch 4 (PS4) Circuit—High	Yes		6-248
P0880	TCM Power Input Signal	No		6-254
P0881	TCM Power Input Signal—Performance	No		6-257
P0882	TCM Power Input Signal—Low	Yes		6-261
P0883	TCM Power Input Signal—High	Yes		6-264
P088B	Transmission Filter Very Deteriorated	Yes		6-267
P0897	Transmission Fluid Deteriorated	Yes		6-269
P0960	Main Pressure Modulation Solenoid Control Circuit Open	Yes		6-271
P0961	Main Pressure Modulation Solenoid System Performance	Yes		6-276
P0962	Main Pressure Modulation Solenoid Control Circuit Low	Yes		6-279
P0963	Main Pressure Modulation Solenoid Circuit High	Yes		6-283
P0964	Pressure Control Solenoid 2 (PCS2) Control Circuit—Open	Yes		6-287
P0966	Pressure Control Solenoid 2 (PCS2) Control Circuit—Low	Yes		6-292
P0967	Pressure Control Solenoid 2 (PCS2) Control Circuit—High	Yes		6-296
P0968	Pressure Control Solenoid 3 (PCS3) Control Circuit Open	Yes		6-300
P0970	Pressure Control Solenoid 3 (PCS3) Control Circuit Low	Yes		6-305
P0971	Pressure Control Solenoid 3 (PCS3) Control Circuit High	Yes		6-309
P097A	Shift Solenoid 1 (SS1) Control Circuit—Open	Yes		6-313
P0973	Shift Solenoid 1 (SS1) Control Circuit—Low	Yes		6-318
P0974	Shift Solenoid 1 (SS1) Control Circuit—High	Yes		6-322
P097B	Shift Solenoid 2 (SS2) Control Circuit—Open	Yes		6-326
P0976	Shift Solenoid 2 (SS2) Control Circuit—Low	Yes		6-331
P0977	Shift Solenoid 2 (SS2) Control Circuit—High	Yes		6-335
P097C	Shift Solenoid 3 (SS3) Control Circuit—Open	Yes		6-339

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Table 6–12. Diagnostic Trouble Codes and Descriptions *(cont'd)*

DTC	Description	CHECK TRANS Light	Diagnostic Response	Page
P0979	Shift Solenoid 3 (SS3) Control Circuit—Low	Yes		6-344
P0980	Shift Solenoid 3 (SS3) Control Circuit—High	Yes		6-348
P1790	Gear Shift Module 1 Calibration Invalid	Yes	Shift selector language or units incorrect	6-352
P1891	Throttle Position Sensor (TPS) PWM Signal—Low Input	No		6-353
P1892	Throttle Position Sensor (TPS) PWM Signal—High Input	No		6-356
P2637	Torque Management Feedback Signal—SEM	Yes		6-359
P2641	Torque Management Feedback Signal—LRTP	Yes		6-362
P2669	Actuator Supply Voltage 2 Open	Yes		6-365
P2670	Actuator Supply Voltage 2 (HSD2)—Low	Yes		6-370
P2671	Actuator Supply Voltage 2 (HSD2)—High (Batt)	Yes		6-374
P2684	Actuator Supply Circuit Voltage 3 Open (HSD3)	Yes		6-379
P2685	Actuator Supply Circuit Voltage 3 Low (HSD3)	Yes		6-383
P2686	Actuator Supply Circuit Voltage 3 High (HSD3)	Yes		6-387
P2723	Pressure Control Solenoid 1 (PCS1)—Stuck Off	Yes		6-392
P2724	Pressure Control Solenoid 1 (PCS1)—Stuck On	Yes		6-396
P2727	Pressure Control Solenoid 1 (PCS1) Control Circuit—Open	Yes		6-400
P2729	Pressure Control Solenoid 1 (PCS1) Control Circuit—Low	Yes		6-405
P2730	Pressure Control Solenoid 1 (PCS1) Control Circuit—High	Yes		6-409
P2761	TCC PCS Control Circuit Open	Yes		6-413
P2763	TCC PCS Control Circuit High	Yes		6-418
P2764	TCC PCS Control Circuit Low	Yes		6-422
P2789	Clutch Adaptive Learning at Limit	Yes		6-426
U0073	CAN Communication Bus 1 Off	No		6-428
U0074	CAN Communications BUS 2 Off	No		6-431
U0100	Lost Communication With ECM A	No		6-434

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